



IGNITION & THERMAL BARRIER COATING

FireShell F10E

Water-Based Intumescent Fire Protective Coating

Manufactured by **ICP Construction / ICP Building Solutions Group** · 150 Dascomb Road, Andover, Massachusetts 01810 ·
Country of origin: **United States**

Building code requires that spray foam be separated from living space by a 15-minute thermal barrier — normally half-inch drywall. Where drywall is impractical, such as an exposed attic or crawl space, an approved intumescent coating does the same job. FireShell F10E is a water-based, single-component coating that expands in a fire to form a protective char layer, starving the fire of oxygen and keeping flame off the foam beneath.

< 10 g/L

VOC content
EPA low-VOC

Class A

fire rating
ASTM E84

ESR-3997

ICC-ES evaluation
Code approved

No growth

mold & fungus
ASTM D3278

WHY WE SPECIFY IT

1 Code-compliant without drywall

F10E complies with International Building Code requirements as an alternative thermal barrier coating over most SPF insulation systems under ICC-ES ESR-3997 — meeting IBC 2603.9 and IRC R316.6 over spray foam.

2 Non-toxic and non-fuming

Water-based, single-component, and non-flammable. It is formaldehyde-free, halogen-free, HFC-free and PBDE-free, and carries LEED Certified Green Product status and EPA low-VOC compliance.

3 Mold resistant

Rated 10 with no mold growth under ASTM D3278. In an attic environment that matters as much as the fire rating does.

4 Documented application thickness

We check wet film thickness during application using a wet film gauge on coupons spaced evenly through the work area. That record is how you know the coating was applied at the rate the code approval requires, not just billed for.

HEALTH & INDOOR AIR QUALITY

- **VOC content below 10 g/L.** For comparison, standard interior latex paint typically runs 50–100 g/L. F10E meets EPA low-VOC standards and is a LEED Certified Green Product.
- **Free of the compounds people ask about.** Formaldehyde-free, halogen-free, HFC-free and PBDE-free. Non-toxic and non-fuming.
- **Odor during drying is comparable to low-odor latex paint.** Under proper drying conditions these odors dissipate quickly and do not linger. Full cure is 2–4 weeks, after which the coating is a durable, inert finish.
- **Ventilation is not optional and we treat it that way.** Closed spaces such as unvented attics require mechanical, negative-pressure cross ventilation during spraying and drying. We supply fresh exterior air at one end and exhaust the moisture-laden air to open exterior space at the other, never into a living area or garage. *Skipping this step is the single most common cause of lingering odor complaints on coated attics.*

TECHNICAL PROPERTIES

PROPERTY	VALUE	TEST METHOD
VOC content	< 10 g/L	—
Solids by volume	59%	ASTM D2697
Solids by weight	70%	—
Weight per gallon	11.5 lbs (± 0.3)	ASTM D1475
Mold & fungus resistance	10 — no mold growth	ASTM D3278
Coverage	80 - 115 sq ft per gallon	—
Maximum single-pass rate	27 wet mils	—
Tack-free time	60 - 90 min at 77°F / 50% RH	—
Dry / recoat time	2 - 3 hours at 77°F / 50% RH	—
Full cure	2 - 4 weeks	—
Colors available	White, Gray, Charcoal Black	—
Shelf life	1 year	—

CODE COMPLIANCE & CERTIFICATIONS

ICC-ES evaluation report	ESR-3997 — alternative thermal barrier coating over most SPF systems
Interior surfaces	Meets IBC 803.1.2.1 and IRC R302.9.4
Over spray polyurethane foam	Meets IBC 2603.9 and IRC R316.6
Surface burning	ASTM E84 — Class A
Fire-retardant paint listing	FM 4975 — listed over combustible surfaces
Environmental	LEED Certified Green Product; meets EPA low-VOC standard

ENVIRONMENTAL & CONTENT

Formaldehyde None	Technical Data Sheet (TDS)	fireshellcoatings.com → Technical Data
Halogens None	Safety Data Sheet (SDS)	fireshellcoatings.com → Safety Data Sheets
HFC None	ICC-ES Evaluation Report	ESR-3997 — icc-es.org
PBDE None	Full TDS and SDS for every product on your project are available on request, at no charge, same day.	

COATING & ASSEMBLY NOTE

Coverage rate is product-specific. Every SPF insulation system has its own approved application rate under the F10E product matrix and ICC-ES ESR-3997 — a coating must be full-scale tested over the exact foam it is applied to. We confirm the correct rate against the report for the foam installed on your project before we coat.